

LAFAYETTE HILL, PA, USA

WMO: 720304

Lat: 40.100N

Lon: 75.267W

Elev: 302

StdP: 14.54

Time Zone: -5.00 (NAE)

Period: 04-19

WBAN: 64752

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	11.9	16.1	-4.3	4.4	16.1	0.5	5.7	19.8	18.8	39.0	16.9	35.6	4.1	300	0.420

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.4	90.9	74.0	88.6	73.0	86.3	71.8	77.3	86.3	75.8	84.3	74.4	82.3	6.4	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
74.5	130.7	81.3	73.0	124.0	79.5	72.3	120.8	78.7	40.7	85.6	39.3	84.5	38.2	82.8	82.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
16.6	13.9	12.0	DB	5.6	95.8	4.7	3.8	2.2	98.5	-0.5	100.7	-3.1	102.8	-6.5	105.5
			WB	4.3	80.2	4.7	1.1	0.9	81.0	-1.8	81.6	-4.4	82.3	-7.9	83.1

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	53.8	31.9	33.8	41.5	52.5	62.1	70.5	75.2	73.1	66.6	55.1	44.4	37.2
	DBStd	16.99	9.59	8.93	9.14	8.27	8.04	6.15	5.04	5.15	6.66	8.19	7.96	8.78
	HDD50	2040	565	460	293	62	5	0	0	0	0	41	205	409
	HDD65	5060	1027	874	729	386	156	21	1	3	62	322	618	861
	CDD50	3413	3	6	30	137	379	616	781	716	497	199	36	13
	CDD65	958	0	0	0	11	65	187	317	254	109	15	0	0
	CDH74	8880	0	0	11	148	658	1719	3049	2251	938	105	1	0
	CDH80	2918	0	0	0	34	177	541	1175	717	259	15	0	0
Wind	WSAvg	4.3	4.9	5.2	5.5	5.6	4.4	4.0	3.2	2.9	3.2	3.7	4.2	4.6
Precipitation	PrecAvg	46.7	3.6	2.6	4.3	3.8	3.6	4.0	4.5	4.3	4.5	4.2	3.2	4.1
	PrecMax	62.6	6.1	5.7	6.9	9.8	7.6	10.7	10.1	9.4	11.8	9.2	5.3	9.3
	PrecMin	34.8	1.5	0.6	1.1	1.2	1.2	1.5	1.1	1.5	0.3	0.3	1.1	1.1
	PrecStd	6.9	1.3	1.3	1.6	1.9	1.4	2.2	2.0	1.9	2.8	2.4	1.2	1.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	63.0	66.2	75.0	85.7	88.5	93.0	96.6	93.3	90.4	81.6	71.6	66.4
		MCWB	57.1	57.7	58.9	64.5	71.3	73.2	75.3	72.1	73.7	68.5	61.5	62.5
	2%	DB	56.9	59.1	67.9	78.8	85.1	89.7	91.2	89.8	86.4	76.9	66.5	61.1
		MCWB	52.4	51.8	55.9	61.3	69.3	72.7	74.9	74.1	71.4	66.4	59.4	57.9
	5%	DB	51.9	53.6	61.4	72.9	81.4	86.1	89.6	86.4	82.2	72.7	62.9	55.0
		MCWB	47.8	47.0	51.6	58.5	67.3	71.3	74.2	71.9	69.2	64.8	55.8	51.6
	10%	DB	45.8	46.3	56.2	68.5	77.4	82.2	86.5	83.9	79.4	69.9	58.8	51.5
		MCWB	41.9	41.0	48.8	56.6	65.5	68.9	72.5	71.0	68.0	62.8	52.5	47.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	60.5	60.3	62.3	67.3	74.3	76.9	80.0	78.9	77.3	72.5	65.1	63.3
		MCDB	61.8	64.3	70.9	76.2	84.5	88.2	91.4	86.7	85.3	77.8	67.6	64.9
	2%	WB	53.5	53.4	57.8	64.4	72.2	74.9	77.7	76.9	75.1	70.1	60.4	58.4
		MCDB	55.3	57.0	64.9	74.6	81.7	84.7	87.3	84.7	82.7	73.9	64.7	61.3
	5%	WB	47.3	46.4	53.7	61.6	70.0	73.3	76.1	75.2	73.2	67.3	57.2	53.3
		MCDB	51.6	51.5	60.0	70.3	77.2	82.4	85.1	82.1	79.3	71.8	62.0	55.0
	10%	WB	42.4	42.1	49.5	58.4	67.5	71.7	74.7	73.5	71.4	63.8	53.9	47.3
		MCDB	46.2	46.5	55.9	66.4	75.0	79.7	83.0	79.9	76.8	66.9	57.4	50.0
Mean Daily Temperature Range	5% DB	MDBR	14.7	15.5	17.3	21.0	20.6	19.7	19.4	19.3	20.5	19.6	18.1	14.1
		MCDBR	22.4	23.9	26.0	27.9	25.0	23.2	22.1	21.5	23.8	23.9	24.0	20.0
		MCWBR	18.6	18.2	16.6	13.9	11.6	9.4	8.0	8.8	10.9	13.8	16.8	17.5
	5% WB	MCDBR	20.8	22.5	23.1	22.7	21.8	20.0	18.6	17.1	19.2	18.1	20.6	18.5
		MCWBR	18.6	18.9	17.3	14.7	11.9	9.3	8.0	8.4	10.6	12.8	17.0	18.3
Clear-Sky Solar Irradiance	taub	0.309	0.320	0.360	0.391	0.437	0.497	0.498	0.492	0.405	0.358	0.325	0.307	
	taud	2.455	2.412	2.368	2.308	2.208	2.083	2.094	2.097	2.310	2.453	2.511	2.511	
	Ebn at Noon	272	285	283	280	268	252	250	247	264	267	264	264	
	Edh at Noon	25	30	35	40	45	51	50	49	37	29	24	22	
All-Sky Solar Radiation	RadAvg	602	889	1174	1522	1697	1856	1895	1647	1364	970	687	520	
	RadStd	52	98	93	135	201	150	91	105	156	102	61	54	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (75 neighbors)	+0.74	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+218	+122

Nomenclature: See separate page